Sections: B,C & D

Linear Algebra (MT-1004)

Homework#21

Instructor: Muhammad Ali

Question 1

Find the limiting values of y_k and z_k, $(k \to \infty)$ if

$$\begin{aligned} y_{k+1} &= .8y_k + .3z_k, \quad y_0 = 0 \\ z_{k+1} &= .2y_k + .7z_k, \quad z_0 = 5. \end{aligned}$$

Question 2

Solve the recurrence relation with the given initial conditions.

- $a_0 = 4, a_1 = 1, a_n = a_{n-1} - \frac{a_{n-2}}{4}$, for $n \geq 2$.
- $a_n = 2a_{n-1} + a_{n-2} - 2a_{n-3}$, subject to $a_0 = 2, a_1 = 2, a_2 = 4$, for $n \geq 3$.